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Class: Signals and Systems

Lab #: 05

Date: October 2, 2025

Lab 4: Frequency Responses

Solutions for Tasks 1–9

```
% Difference equation:
%  $y(n) = 0.155x(n) + 0.096x(n-1) + 0.155x(n-2) + 1.131y(n-1) - 0.64y(n-2)$ 

clc; clear; close all;
```

1. Transfer Function $H(z)$

```
fprintf('STEP-BY-STEP DERIVATION OF H(z) FROM THE DIFFERENCE EQUATION\n\n');
```

STEP-BY-STEP DERIVATION OF H(z) FROM THE DIFFERENCE EQUATION

```
fprintf('Given difference equation:\n');
```

Given difference equation:

```
fprintf('  $y(n) = 0.155 x(n) + 0.096 x(n-1) + 0.155 x(n-2) + 1.131 y(n-1) - 0.64$   
 $y(n-2)$ \n\n');
```

$$y(n) = 0.155 x(n) + 0.096 x(n-1) + 0.155 x(n-2) + 1.131 y(n-1) - 0.64 y(n-2)$$

```
fprintf('Apply Z-transform (assume zero initial conditions):\n');
```

Apply Z-transform (assume zero initial conditions):

```
fprintf('  $Y(z) = 0.155 X(z) + 0.096 z^{-1} X(z) + 0.155 z^{-2} X(z) + 1.131 z^{-1} Y(z) -$   
 $0.64 z^{-2} Y(z)$ \n\n');
```

$$Y(z) = 0.155 X(z) + 0.096 z^{-1} X(z) + 0.155 z^{-2} X(z) + 1.131 z^{-1} Y(z) - 0.64 z^{-2} Y(z)$$

```
fprintf('Move ALL terms with Y(z) to the left-hand side:\n');
```

Move ALL terms with $Y(z)$ to the left-hand side:

```
fprintf(' Y(z) - 1.131 z^-1 Y(z) + 0.64 z^-2 Y(z) = (0.155 + 0.096 z^-1 + 0.155 z^-2) X(z)\n\n');
```

$$Y(z) - 1.131 z^{-1} Y(z) + 0.64 z^{-2} Y(z) = (0.155 + 0.096 z^{-1} + 0.155 z^{-2}) X(z)$$

```
fprintf('Factor Y(z) from the left-hand side:\n');
```

Factor $Y(z)$ from the left-hand side:

```
fprintf(' Y(z) * (1 - 1.131 z^-1 + 0.64 z^-2) = B(z) * X(z)\n');
```

$$Y(z) * (1 - 1.131 z^{-1} + 0.64 z^{-2}) = B(z) * X(z)$$

```
fprintf(' where B(z) = 0.155 + 0.096 z^-1 + 0.155 z^-2\n\n');
```

$$\text{where } B(z) = 0.155 + 0.096 z^{-1} + 0.155 z^{-2}$$

```
fprintf('Divide both sides by A(z) and X(z) to obtain the transfer function:\n');
```

Divide both sides by $A(z)$ and $X(z)$ to obtain the transfer function:

```
fprintf(' H(z) = Y(z)/X(z) = B(z) / A(z)\n\n');
```

$$H(z) = Y(z)/X(z) = B(z) / A(z)$$

```
fprintf(' with\n');
```

with

```
fprintf(' A(z) = 1 - 1.131 z^-1 + 0.64 z^-2\n');
```

$$A(z) = 1 - 1.131 z^{-1} + 0.64 z^{-2}$$

```
fprintf(' B(z) = 0.155 + 0.096 z^-1 + 0.155 z^-2\n\n');
```

$$B(z) = 0.155 + 0.096 z^{-1} + 0.155 z^{-2}$$

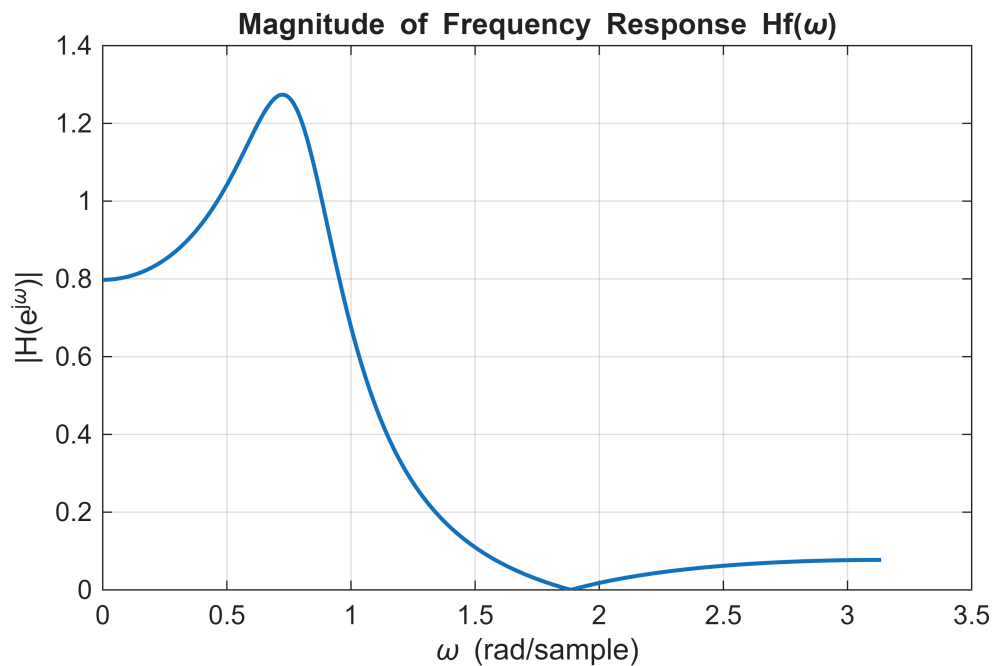
2. Magnitude of Frequency Response $H_f(w)$

```
b = [0.155 0.096 0.155]; % Numerator coefficients
a = [1 -1.131 0.64]; % Denominator coefficients

[H,w] = freqz(b,a);

figure;
plot(w,abs(H),'LineWidth',1.5);
title('Magnitude of Frequency Response Hf(\omega)');
xlabel('\omega (rad/sample)'); ylabel('|H(e^{j\omega})|');
```

```
grid on;
```



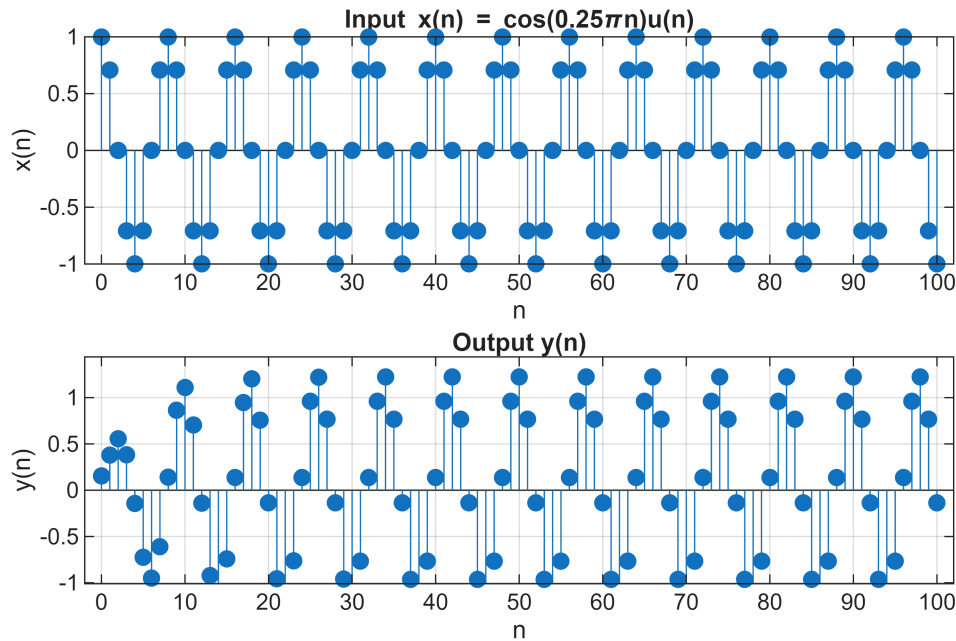
3. Input $x(n) = \cos(0.25\pi n)u(n)$, Output $y(n)$

```
n = 0:100;
x = cos(0.25*pi*n).*(n>=0);
y = filter(b,a,x);

figure;
subplot(2,1,1);
stem(n,x,'filled');
title('Input x(n) = cos(0.25\pin)u(n)');
xlabel('n'); ylabel('x(n)'); grid on;

subplot(2,1,2);
stem(n,y,'filled');
title('Output y(n)');
xlabel('n'); ylabel('y(n)'); grid on;

subplot(2,1,2)
xlim([-2.0 102.0])
ylim([-1.01 1.44])
```



```
% Display information about the input signal
disp('Input x(n) = cos(0.25*pi*n)u(n):');
```

```
Input x(n) = cos(0.25*pi*n)u(n):
```

```
disp('The input is a cosine wave starting at n=0, with frequency 0.25*pi.');
```

```
The input is a cosine wave starting at n=0, with frequency 0.25*pi.
```

```
disp('This wave is multiplied by the unit step function u(n), making it zero for n<0.');
```

```
This wave is multiplied by the unit step function u(n), making it zero for n<0.
```

```
% Display information about the output signal
disp('Output y(n):');
```

```
Output y(n):
```

```
disp('The output is the result of filtering the input signal x(n) through the system.');
```

```
The output is the result of filtering the input signal x(n) through the system.
```

```
disp('The filter affects the amplitude and phase of the frequency components, altering the input signal, but maintains the same frequency This is typical of impulse responses to frequency dependent inputs.');
```

```
The filter affects the amplitude and phase of the frequency components, altering the input signal, but maintains the same frequency.
```

```
% Comment on the frequency response
```

```
disp('Prediction based on frequency response:');
```

Prediction based on frequency response:

```
disp('The filters frequency response at the frequency 0.25*pi will determine the  
amplitude scaling and phase shift of the output.');
```

The filters frequency response at the frequency 0.25π will determine the amplitude scaling and phase shift of the

```
disp('By analyzing Hf(0.25*pi), we can predict how the output y(n) differs from the  
input x(n) due to scaling and phase shift but has the same frequency');
```

By analyzing $H_f(0.25\pi)$, we can predict how the output $y(n)$ differs from the input $x(n)$ due to scaling and phase shift

4. Exact Value of $H_f(0.25\pi)$

```
omega0 = 0.25*pi;  
z = exp(1j*omega0);  
Bz = polyval(b,z);  
Az = polyval(a,z);  
H_eval = Bz/Az;  
disp('Hf(0.25π) = ');
```

$H_f(0.25\pi) =$

```
disp('The exact value of Hf(w) at w = 0.25*pi is: ')
```

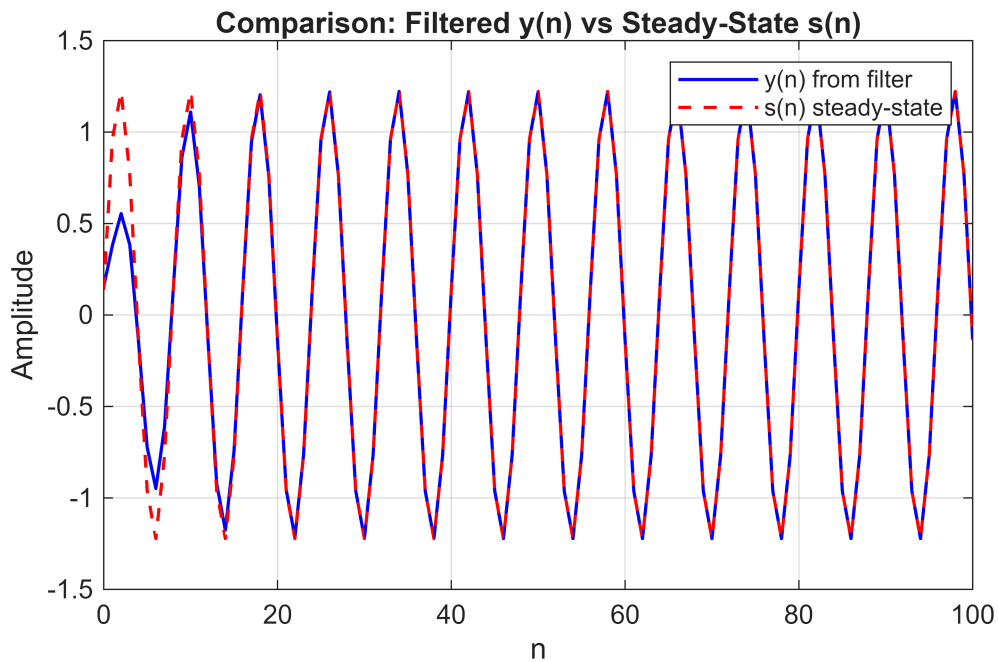
The exact value of $H_f(w)$ at $w = 0.25\pi$ is:

```
disp(H_eval);
```

$0.1376 - 1.2227i$

5. Steady-State Signal

```
R = abs(H_eval);  
phi = angle(H_eval);  
s = R*cos(0.25*pi*n + phi);  
  
figure;  
plot(n,y,'b','LineWidth',1.2); hold on;  
plot(n,s,'r--','LineWidth',1.2);  
xlabel('n'); ylabel('Amplitude');  
title('Comparison: Filtered y(n) vs Steady-State s(n)');  
legend('y(n) from filter','s(n) steady-state');  
grid on;
```



```
disp('The plots agree after teh transients die out.')
```

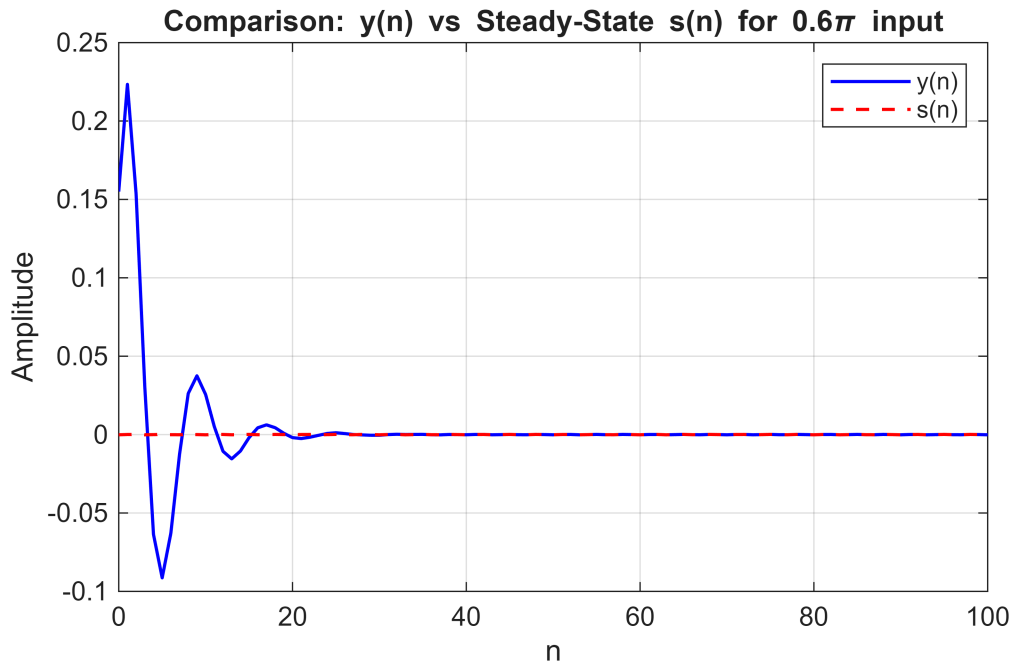
The plots agree after teh transients die out.

6. Input $x(n) = \cos(0.6\pi n)u(n)$

```
x2 = cos(0.6*pi*n).*(n>=0);
y2 = filter(b,a,x2);

omega1 = 0.6*pi;
z1 = exp(1j*omega1);
H_eval2 = polyval(b,z1)/polyval(a,z1);
R2 = abs(H_eval2);
phi2 = angle(H_eval2);
s2 = R2*cos(0.6*pi*n + phi2);

figure;
plot(n,y2,'b','LineWidth',1.2); hold on;
plot(n,s2,'r--','LineWidth',1.2);
xlabel('n'); ylabel('Amplitude');
title('Comparison: y(n) vs Steady-State s(n) for 0.6\pi input');
legend('y(n)','s(n)');
grid on;
```



```
% Display information about the output signal y2
disp('Output y2(n):');
```

Output y2(n):

```
disp('The output y2(n) is the result of filtering the input signal x2(n) through
the system.');
```

The output y2(n) is the result of filtering the input signal x2(n) through the system.

```
disp('The filter alters the amplitude and phase of the frequency component at omega
= 0.6*pi.');
```

The filter alters the amplitude and phase of the frequency component at $\omega = 0.6\pi$.

```
% Prediction from frequency response H(f)
disp('Prediction of y(n) from H(f):');
```

Prediction of y(n) from H(f):

```
disp('The output y(n) can be predicted from the systems frequency response H(f)
evaluated at omega = 0.6*pi.');
```

The output y(n) can be predicted from the systems frequency response H(f) evaluated at $\omega = 0.6\pi$.

```
disp('By evaluating the magnitude |H(0.6*pi)| and phase angle angle(H(0.6*pi)), we
can predict the scaling (R2) and phase shift (phi2) of the input frequency.');
```

By evaluating the magnitude $|H(0.6\pi)|$ and phase angle $\angle(H(0.6\pi))$, we can predict the scaling (R2) and phase

```
disp('The filtered output will be a cosine wave with the same frequency but
modified amplitude and phase compared to the input.');
```

The filtered output will be a cosine wave with the same frequency but modified amplitude and phase compared to the input.

```
% Explanation of Steady-State Signal
disp('Steady-State Signal:');
```

Steady-State Signal:

```
disp('A steady-state signal refers to the long-term behavior of the output once all
transient effects have decayed.');
```

A steady-state signal refers to the long-term behavior of the output once all transient effects have decayed.

```
disp('In this case, s2(n) represents the steady-state response, which is the
filters response to a pure sinusoidal input at frequency 0.6*pi.');
```

In this case, $s_2(n)$ represents the steady-state response, which is the filter's response to a pure sinusoidal input at frequency 0.6π .

```
disp('The steady-state signal is a sinusoidal wave with a constant amplitude and
phase determined by the system's frequency response.');
```

The steady-state signal is a sinusoidal wave with a constant amplitude and phase determined by the system's frequency response.

```
% Comment on the comparison of y2(n) and s2(n)
disp('Comparison between y2(n) and s2(n):');
```

Comparison between $y_2(n)$ and $s_2(n)$:

```
disp('If the system reaches steady-state, the actual output y2(n) should match the
steady-state response s2(n) closely as seen in the figure.');
```

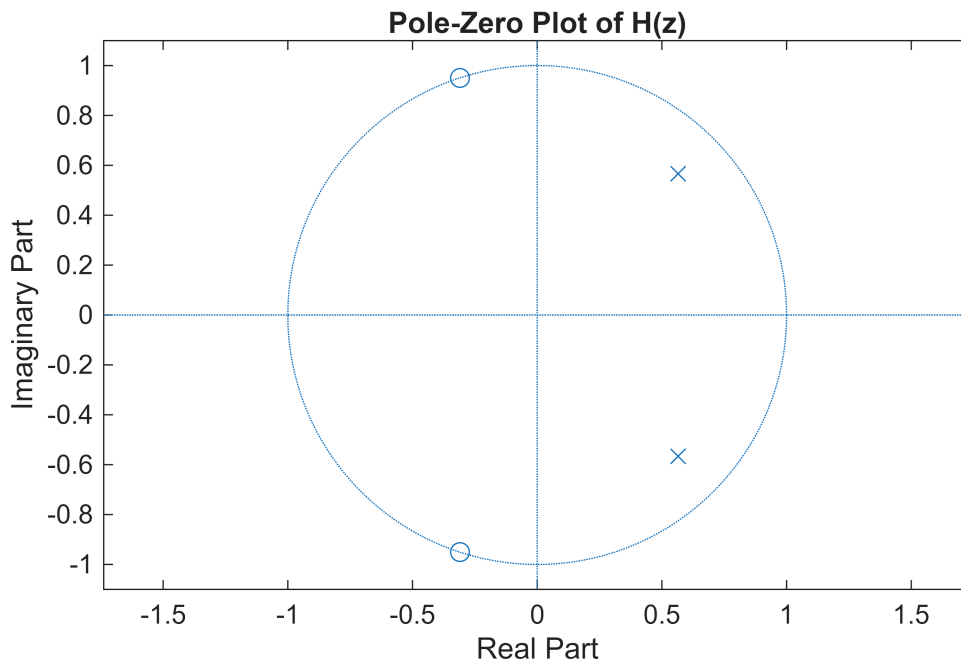
If the system reaches steady-state, the actual output $y_2(n)$ should match the steady-state response $s_2(n)$ closely as seen in the figure.

```
disp('Any differences between y2(n) and s2(n) are due to the transient behavior of
the filter, which will decay over time.');
```

Any differences between $y_2(n)$ and $s_2(n)$ are due to the transient behavior of the filter, which will decay over time.

7. Poles and Zeros

```
figure;
zplane(b,a);
title('Pole-Zero Plot of H(z)');
```

```
disp('Zeros are points where the numerator equals zero.');
```

Zeros are points where the numerator equals zero.

```
disp('When e^{j\omega} is close to a zero, the numerator becomes very small, making the overall response |H(e^{j\omega})| close to zero.');
```

When $e^{j\omega}$ is close to a zero, the numerator becomes very small, making the overall response $|H(e^{j\omega})|$ close to zero.

```
disp('Poles are points where the denominator equals zero.');
```

Poles are points where the denominator equals zero.

```
disp('When e^{j\omega} is close to a pole, the denominator becomes very small, causing |H(e^{j\omega})| to become very large (it can go to infinity if the pole is exactly on the unit circle).');
```

When $e^{j\omega}$ is close to a pole, the denominator becomes very small, causing $|H(e^{j\omega})|$ to become very large (it can go to infinity if the pole is exactly on the unit circle).

8. Some Simple Systems

```
systems = {
    [1], [1 -1.8 0.9];    % System 1
    [1], [1 -1.6 0.72];  % System 2
    [1], [1 0.8];        % System 3
};

for k = 1:length(systems)
```

```

b_sys = systems{k,1};
a_sys = systems{k,2};

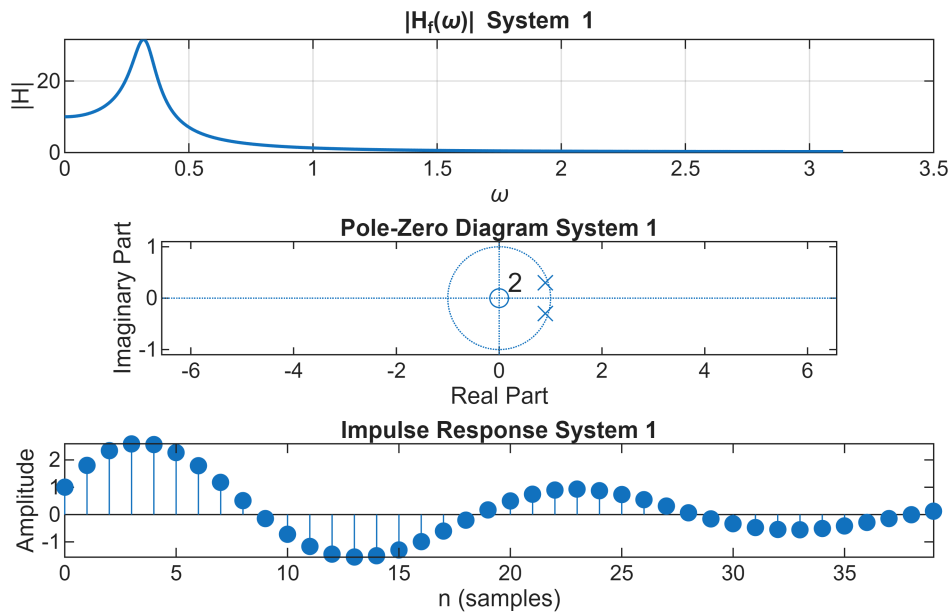
figure;
subplot(3,1,1);
[H_sys,w_sys] = freqz(b_sys,a_sys,512);
plot(w_sys,abs(H_sys),'LineWidth',1.2);
title(['|H_f(\omega)| System ', num2str(k)]);
xlabel('\omega'); ylabel('|H|'); grid on;

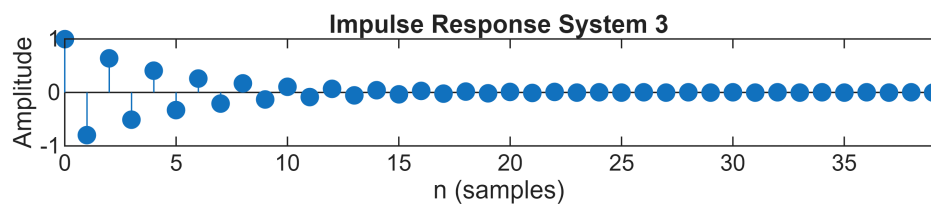
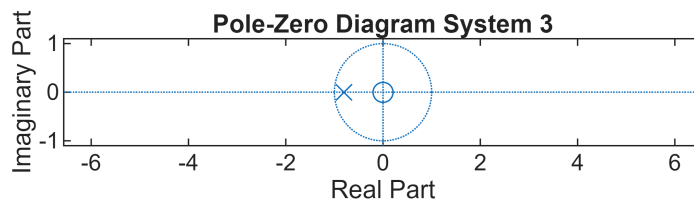
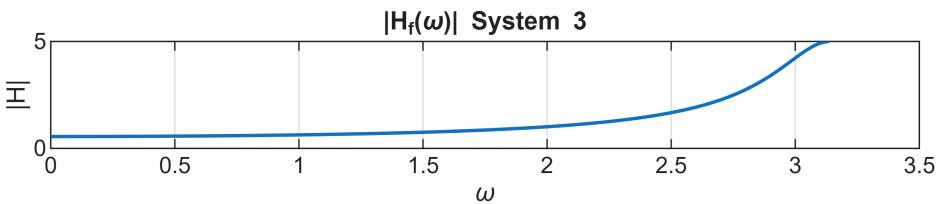
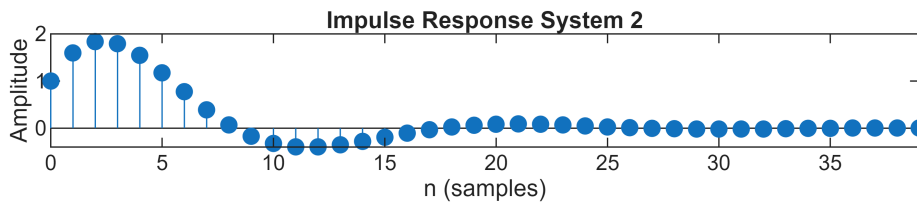
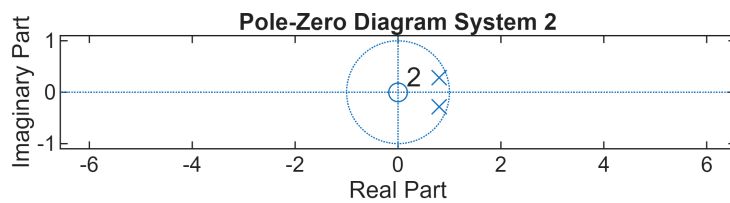
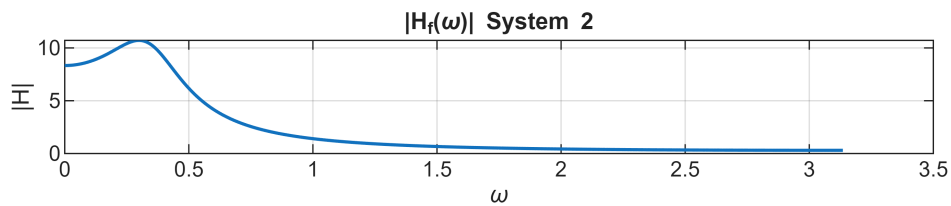
subplot(3,1,2);
zplane(b_sys,a_sys);
title(['Pole-Zero Diagram System ', num2str(k)]);

subplot(3,1,3);
impz(b_sys,a_sys,40);
title(['Impulse Response System ', num2str(k)]);

```

end





```
disp('Observations:');
```

Observations:

```
disp('The shape of the output signal depends on how the input frequency aligns with the poles and zeros of the system.');
```

The shape of the output signal depends on how the input frequency aligns with the poles and zeros of the system.

```
disp('You can predict the frequency response by looking at the pole-zero diagram:');
```

You can predict the frequency response by looking at the pole-zero diagram:

```
disp('- Frequencies near poles (close to the unit circle) will be amplified,  
creating peaks in the response.');
```

- Frequencies near poles (close to the unit circle) will be amplified, creating peaks in the response.

```
disp('- Frequencies near zeros will be attenuated or canceled.');
```

- Frequencies near zeros will be attenuated or canceled.

```
disp('Without detailed computation, you can roughly match diagrams by observing  
where poles and zeros lie relative to the unit circle and the angle corresponding  
to the input frequency.');
```

Without detailed computation, you can roughly match diagrams by observing where poles and zeros lie relative to the